

Maharashtra State Board Of Technical Education, Mumbai

Learning and Assessment Scheme for Post S.S.C Diploma Courses

Programme Name	: Diploma In Computer Technology / Computer Engineering / Computer Science & Engineering / Computer Science																					
Programme Code	: CM / CO / CW / SE										With Effect From Academic Year : 2023-24											
Duration Of Programme	: 6 Semester										Duration : 12 Weeks (Industry) + 10 Weeks (Institute)											
Semester	: Fifth										NCrF Entry Level : 4.0											
											Scheme : K											

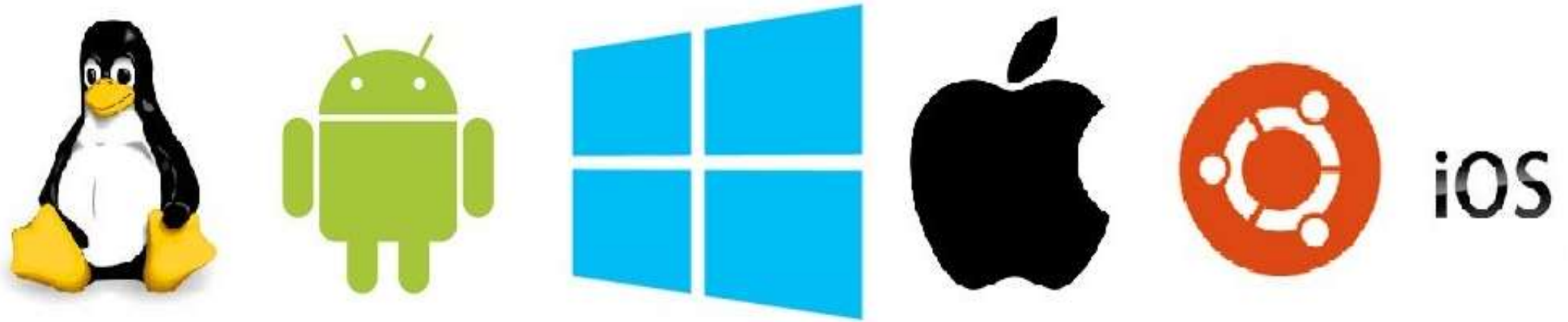
Sr No	Course Title	Abbreviation	Course Type	Course Code	Total IKS Hrs for Sem.	Learning Scheme				Credits	Assessment Scheme												
						Actual Contact Hrs./Week			Self Learning (Activity/ Assignment /Micro Project)		Notional Learning Hrs /Week	Paper Duration (hrs.)	Theory			Based on LL & TL				Based on Self Learning		Total Marks	
						CL	TL	LL								Practical							
													FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
															Max	Min	Max	Min	Max	Min	Max		Min

(All Compulsory)

1	OPERATING SYSTEM	OSY	DSC	315319	-	5	-	2	2	9	3	3	30	70	100	40	25	10	25@	10	25	10	175
2	SOFTWARE ENGINEERING	STE	DSC	315323	-	4	-	4	1	9	3	3	30	70	100	40	25	10	25@	10	25	10	175
3	ENTREPRENEURSHIP DEVELOPMENT AND STARTUPS	ENDS	AEC	315002	-	1	-	2	-	3	1	-	-	-	-	-	50	20	25@	10	-	-	75
4	SEMINAR AND PROJECT INITIATION COURSE	SPI	AEC	315003	-	-	-	1	2	3	1	-	-	-	-	-	25	10	25@	10	25	10	75
5	INTERNSHIP(12 WEEKS)	ITR	INP	315004	-	-	-	-	-	36 - 40	10	-	-	-	-	-	100	40	100#	40	-	-	200

Elective 1 (Any - One)

6	ADVANCE COMPUTER NETWORK	ACN	DSE	315321	-	4	-	2	-	6	2	3	30	70	100	40	25	10	25#	10	-	-	150
	CLOUD COMPUTING	CLC	DSE	315325	-	4	-	2	-	6	2	3	30	70	100	40	25	10	25#	10	-	-	150
	DATA ANALYTICS	DAN	DSE	315326	-	4	-	2	-	6	2	3	30	70	100	40	25	10	25#	10	-	-	150
Total						14		11	5		20		90	210	300		250		225		75		850



Operating System

[315319]

Hamid Ali
Lecturer, Computer Engg.
A.I. ARKP, Panvel.

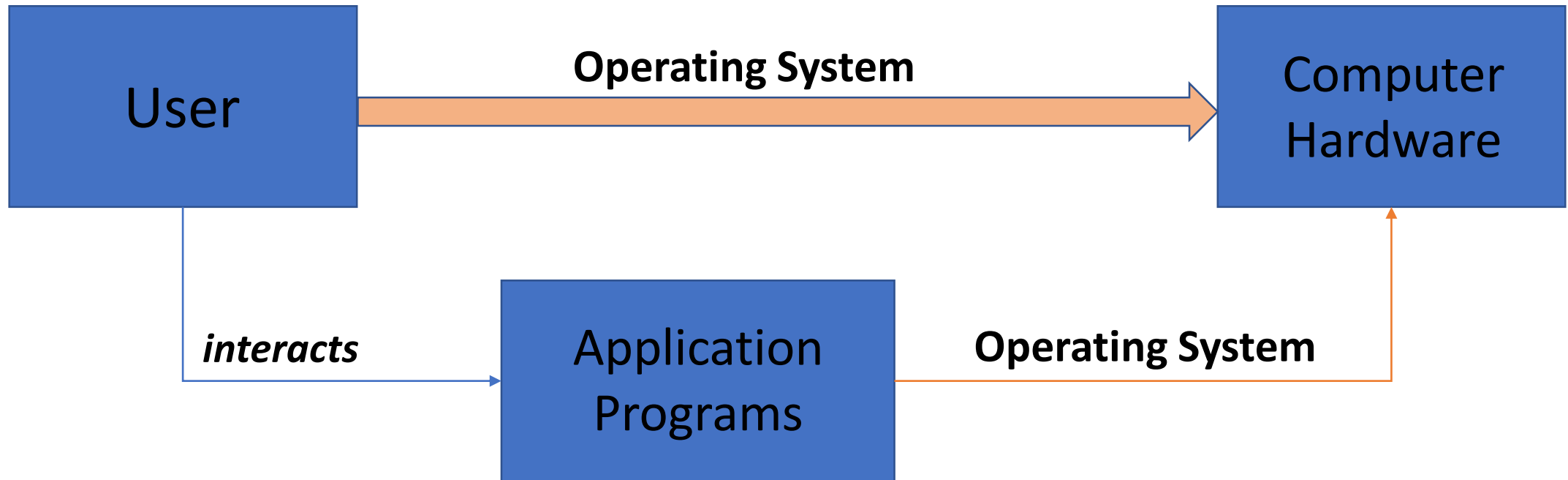
IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme				
				Actual Contact Hrs./Week			SLH	NLH
				CL	TL	LL		
315319	OPERATING SYSTEM	OSY	DSC	5	-	2	2	9

Assessment Scheme											
Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
					Practical						
	FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
	Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
3	30	70	100	40	25	10	25@	10	25	10	175

[Detailed Syllabus:](#)

What is an Operating System ?

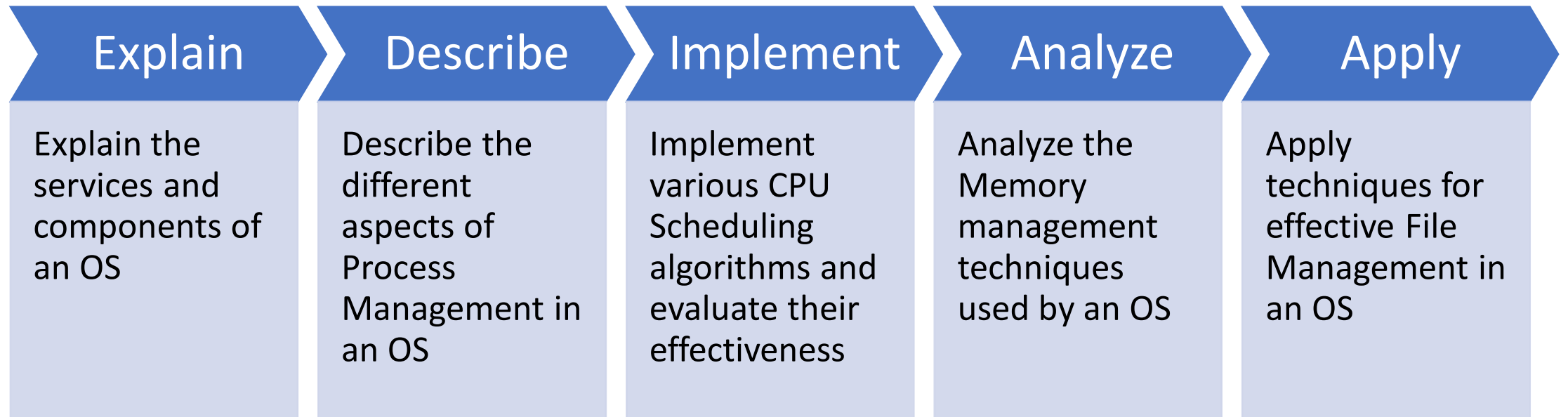


- intermediary between the user and the computer hardware.

Why study Operating System ?

- Education perspective :
 - For Good percentage in diploma
 - For Good percentage in Degree
 - For Good Gate score
- Industry perspective/ Industry Opportunities
 - System Administrator
 - OS Developer
 - Embedded Systems Engineer
 - IT Support Specialist

Course Outcome



Google Classroom code

hwjctfus

Unit 1

Operating System Services and Components

(14 - 16 marks)

Unit Outcome

Describe	Functions of an Operating System.
Explain	Different Services of operating system.
Explain	Use of system call of operating system.
Explain	Activities of operating system in concern with their components.

What is an Operating System?

A program that acts as an intermediary between a user of a computer and the computer hardware



Operating system goals:

Execute user programs and make solving user problems easier

Make the computer system convenient to use

Use the computer hardware in an efficient manner



Microsoft Windows



iOS



UNIX



Ubuntu



BlackBerry OS



Chrome OS



Linux



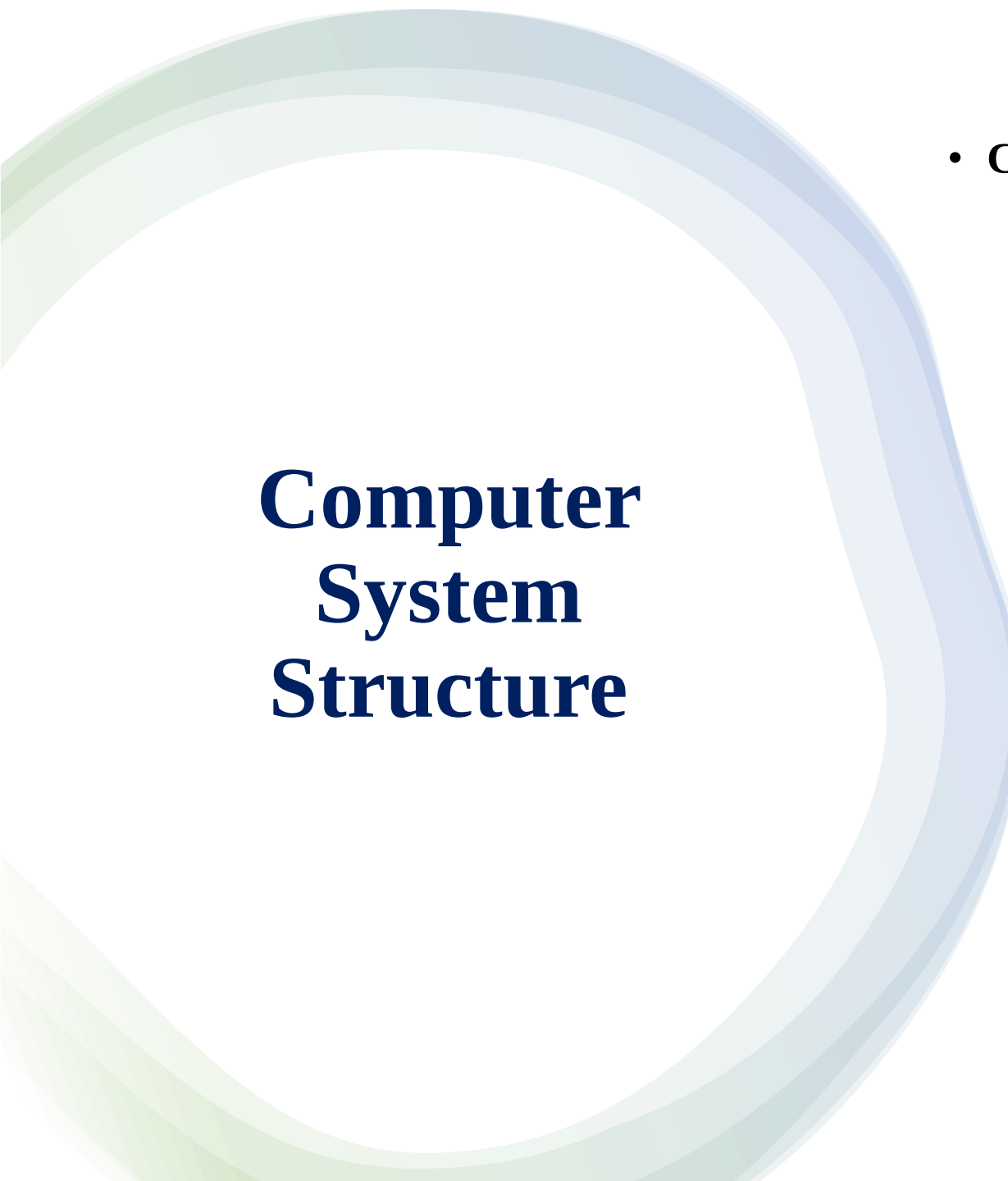
Fedora



Android



macOS



Computer System Structure

- **Computer system can be divided into 4 components:**
 - Hardware – provides basic computing resources
 - CPU, memory, I/O devices
 - Operating system
 - Controls and coordinates use of hardware among various applications and users
 - Application programs – define the ways in which the system resources are used to solve the computing problems of the users
 - Word processors, compilers, web browsers, database systems, video games
 - Users
 - People, machines, other computers

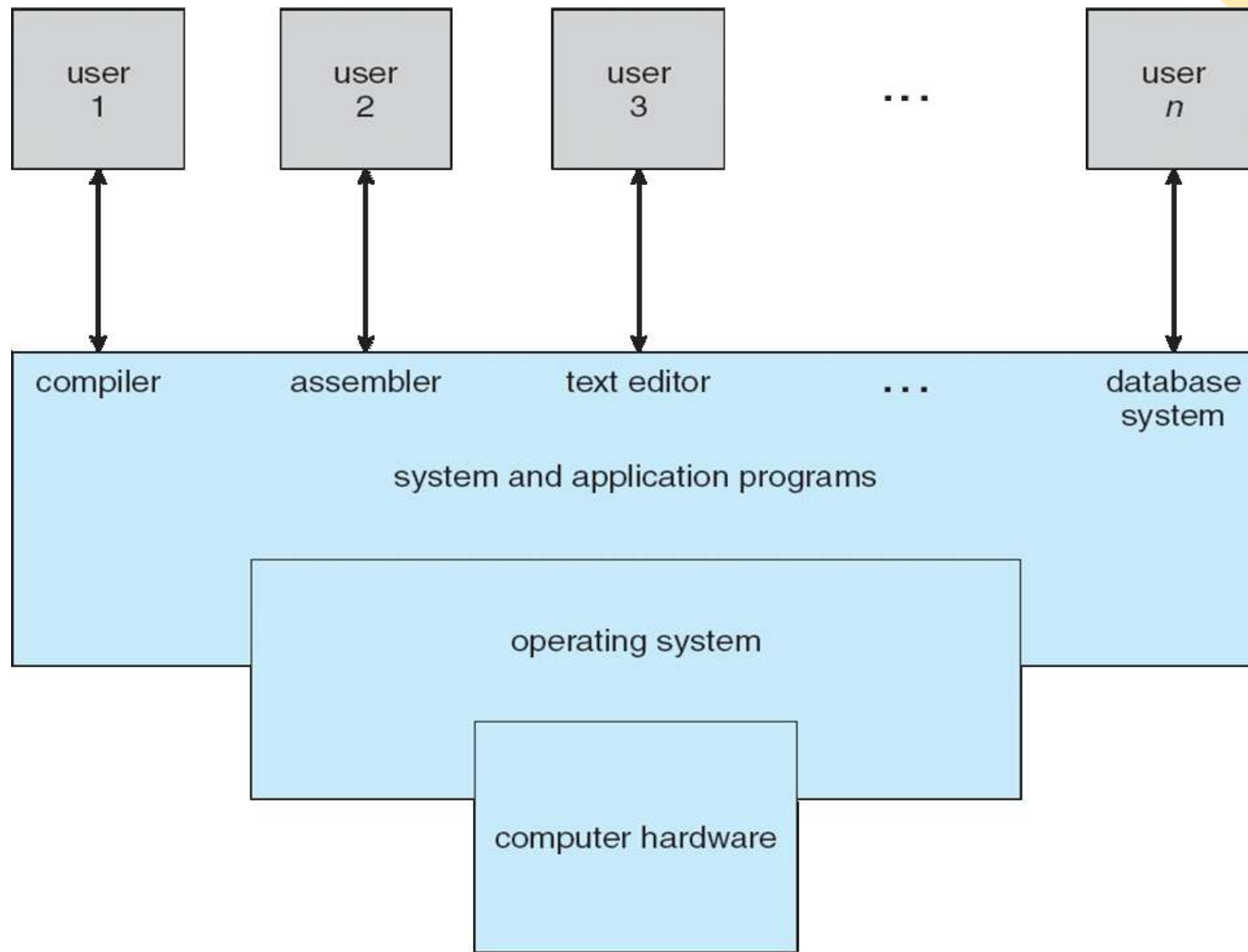


Fig: Components of a Computer System

Functions of OS

- Process Management – Managing the programs that are running.
- Memory Management – Managing and rationing the memory between processes and data.
- Storage Management – Managing the permanent Storage of data on disks or other media
- I/O Management – Managing the input and output
- Device / Resource Management – Managing devices and resources and allowing the users to share the resources
- Security and Protection – Securing the system against possible unauthorized access to data or any other entity. Protecting the parts of the system against damage.
- Booting the System and getting it ready to work.
- Data communications – Providing interface to connect to other computers or allowing others to connect

Types of Operating Systems

Evolutions and Generations of Computer

Types of Operating Systems

Batch
Operating
System

Multi-
Programming
System

Multi-
Processing
System

Multi-Tasking
Operating
System

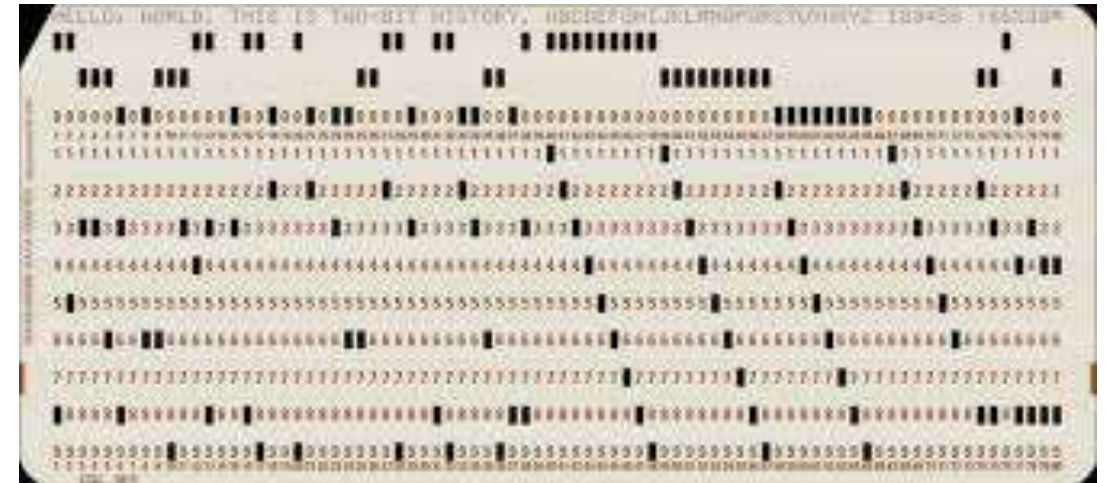
Distributed
Operating
System

Real-Time
Operating
System

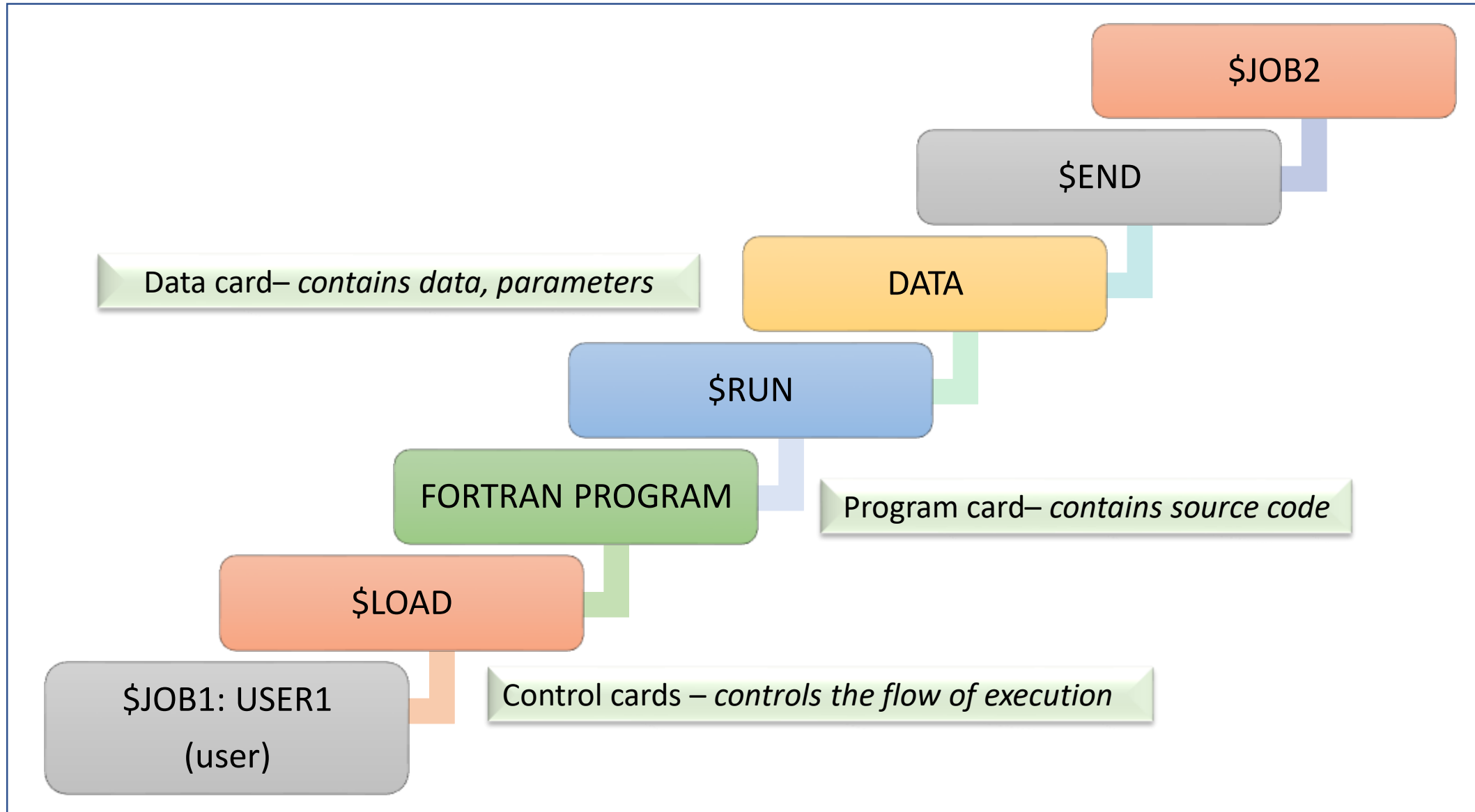
Mobile
Operating
System

Batch OS

- It is the *oldest* of all type.
 - Used during *first and second generation* of computers.
 - Punch cards were used to record the user's job.
-
- A batch stands for the sequence of user jobs. In this system, the job sequence was decided by an operator.



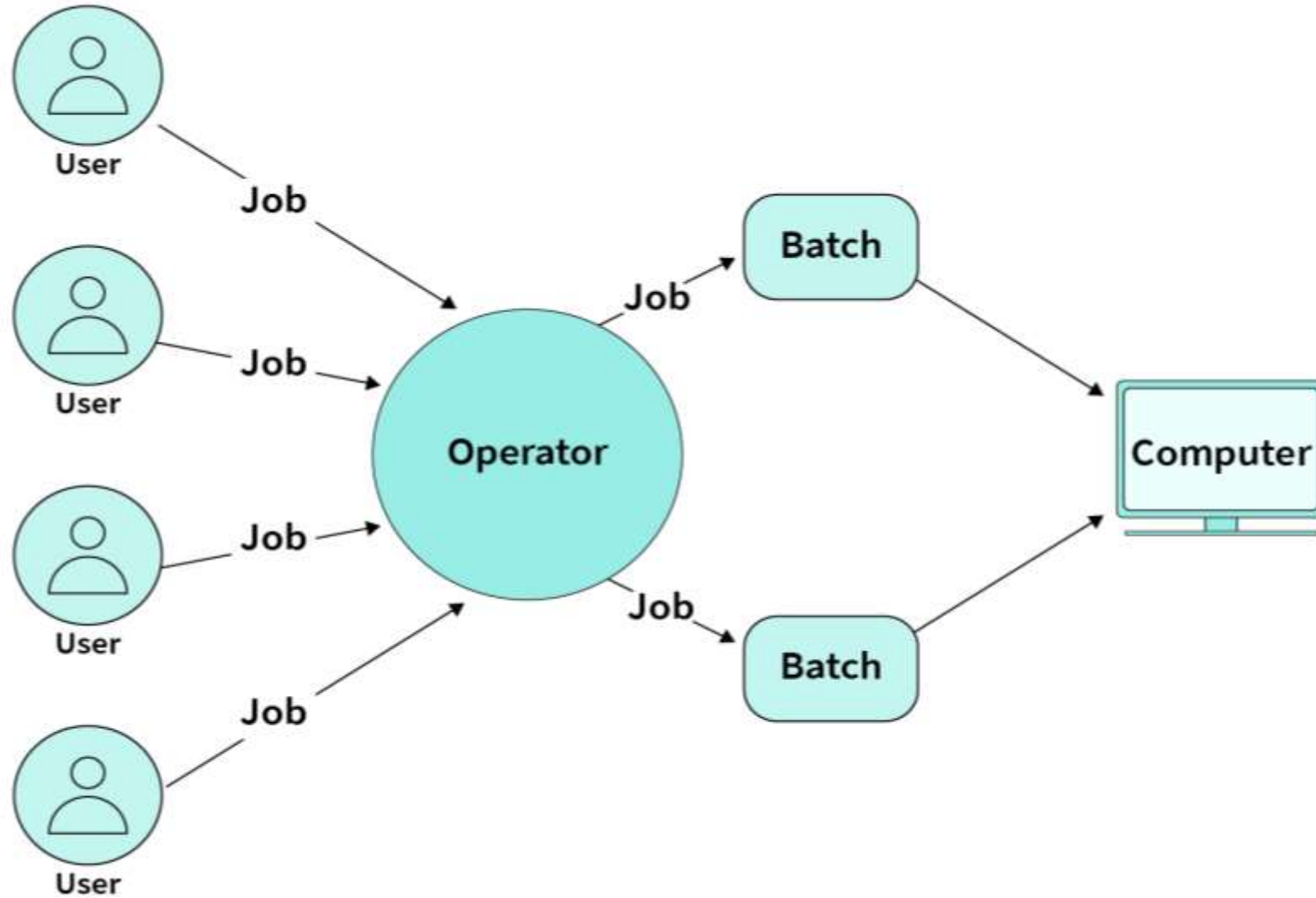
Structure of Batch job



Structure of Batch job

- **\$ JOB:** specifies the job to be done, the user who is doing it.
- **\$ LOAD:** would signify that the object card following it should be loaded in the memory.
- **Program:** It contains executable machine instructions.
- **\$ RUN:** would specify that the program just loaded should be executed.
- **\$ END:** would specify the end of the job.
- **DATA:** would specify the additional data required by the program.

Batch OS

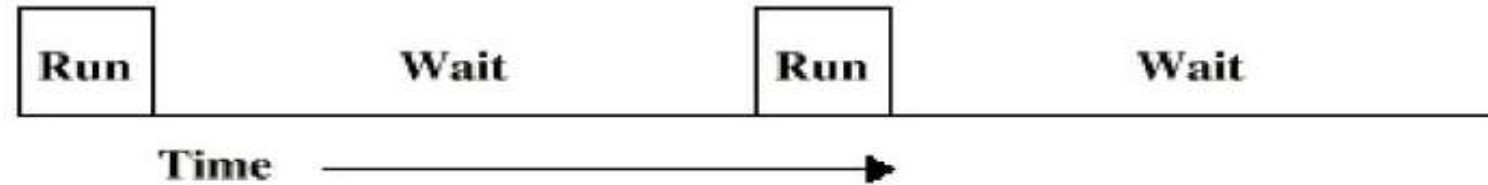


Multi-Programming OS

- In multiprogramming, more than one program lies in the memory.
- The scheduler selects the jobs to be placed in ready queue from a number of programs.
- The ready queue is placed in memory and the existence of more than one program in main memory is known as multiprogramming.
- Since there is **only one processor**, **there multiple programs** cannot be executed at a time.
- Instead the operating system executes part of one program, then the part of another and so on.
- Example of multiprogramming: user can open word, excel, access and other applications in a system.

Multi-Programming OS

- Single program



- Multi-Programming with two programs

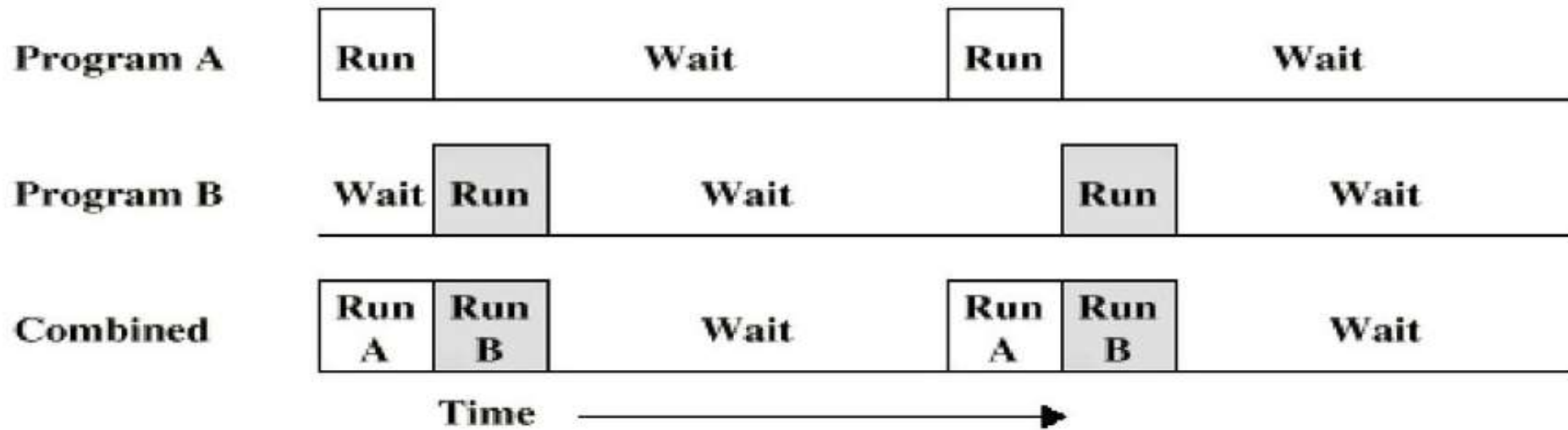
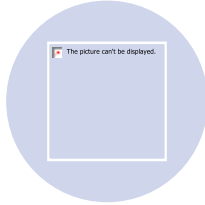
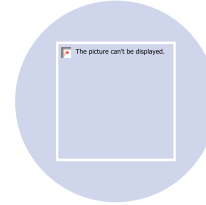


Fig: Multi-programming OS

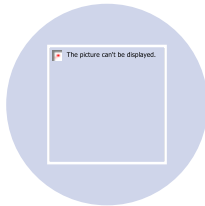
Advantages



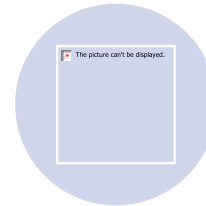
It provides less response time.



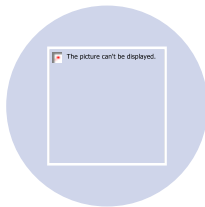
It may help to run various jobs in a single application simultaneously.



It helps to optimize the total job throughput of the computer.

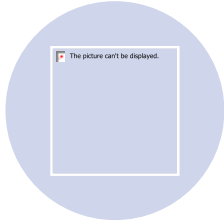


Various users may use the multiprogramming system at once.

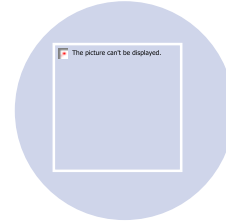


Short-time jobs are done quickly in comparison to long-time jobs.

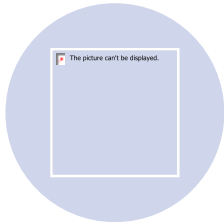
Disadvantages



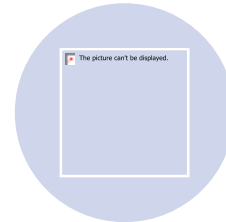
It is highly complicated and sophisticated.



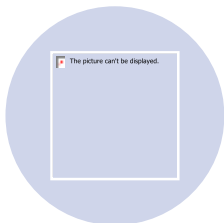
The CPU scheduling is required.



Memory management is needed in the operating system because all types of tasks are stored in the main memory.



The harder task is to handle all processes and tasks.

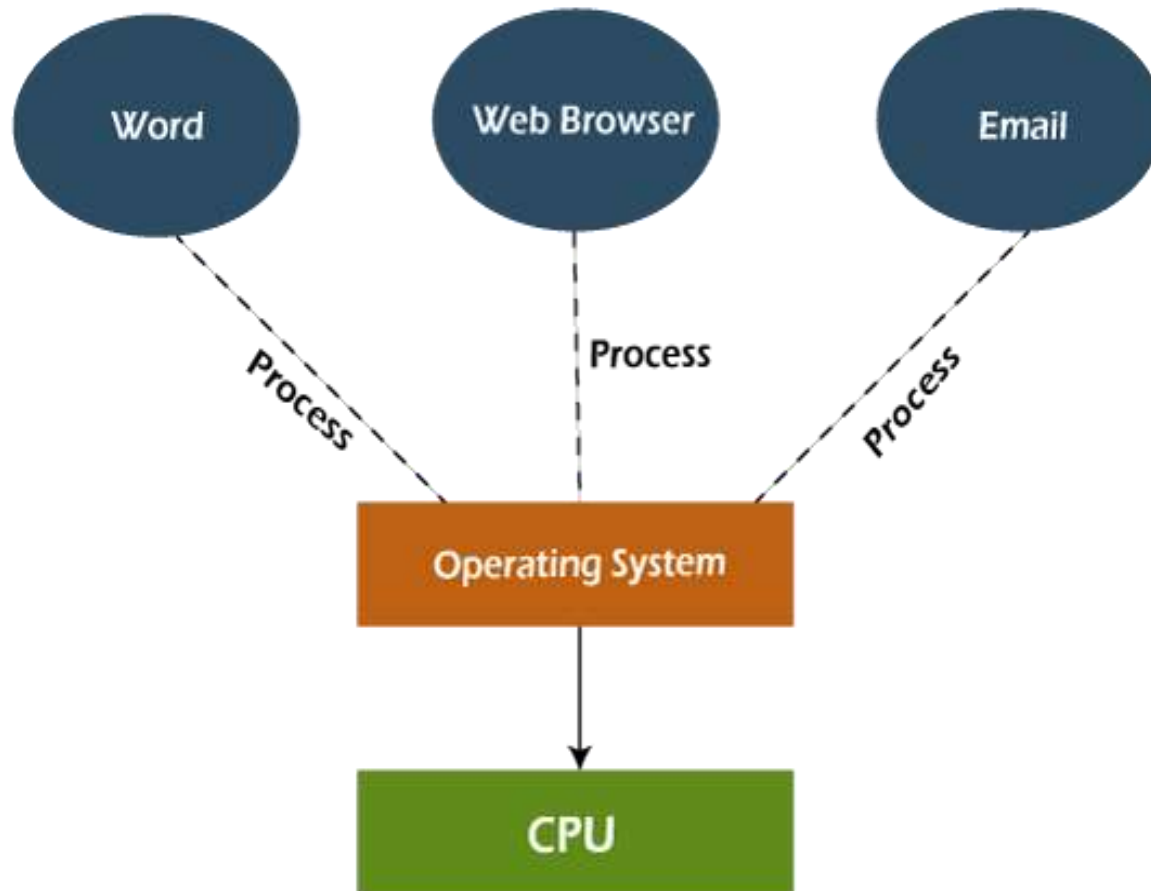


If it has a large number of jobs, then long-term jobs will require a long wait.

Multitasking Operating System

- In time sharing system, the CPU executes multiple jobs by switching among them. The switches occur so frequently that the users can interact with each program while it is running. It **includes an interactive computer system** which provides **direct communication** between the user and the system.
- A time-sharing system allows many users to share the computer resources simultaneously. The time sharing system provides the direct access to many users where CPU time is divided among all the users on scheduled basis.
- The operating **system allocates a time slice to each user**. When this time is expired, it passes control to the next user on the system. The time allowed is extremely small and the users are given the impression that each of them has their own CPU and they are the sole owner of the CPU. In this time slice each user gets attention of the CPU.
- *The **objective** of time-sharing system is to **minimize response time of process**.*

Multitasking Operating System



The multitasking operating system is a logical extension of a multiprogramming system that enables **multiple** programs simultaneously. It allows a user to perform more than one computer task at the same time.

Advantages of Multitasking operating system

- supports multiple users simultaneously.
- It has a well-defined memory management.

Disadvantages of Multitasking operating system

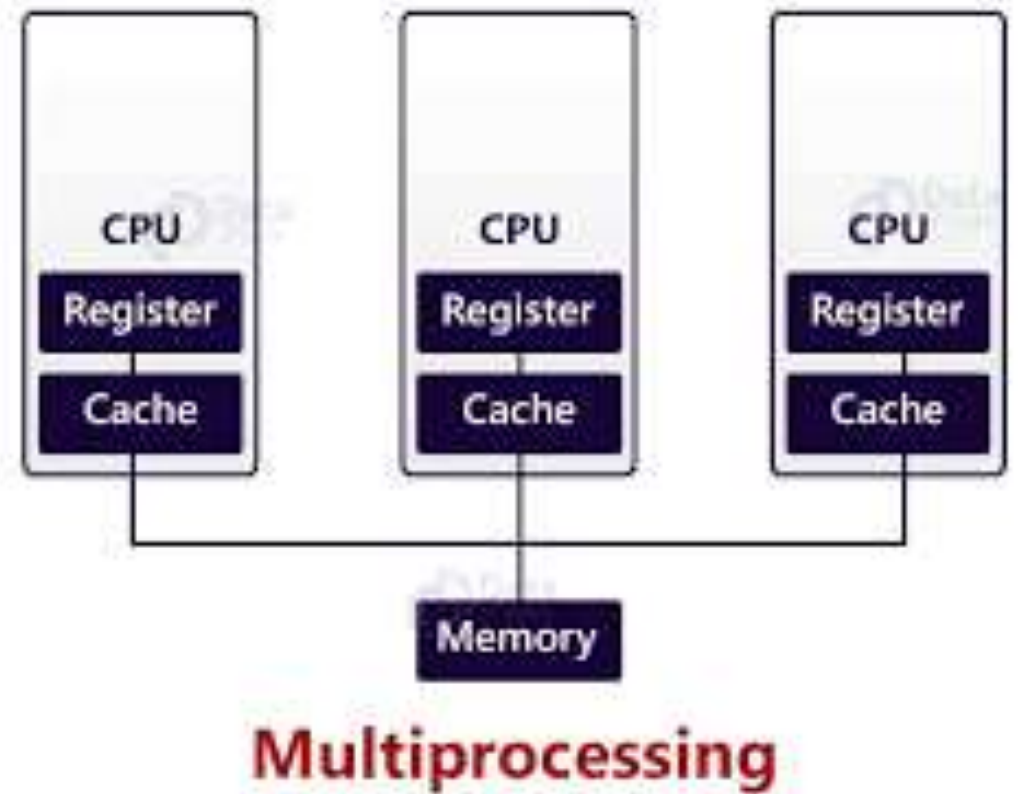
- The multiple processors are busier at the same time to complete any task in a multitasking environment, so the CPU generates more heat.

Differentiate between multitasking and multiprogramming.

Multiprogramming	Multi-tasking
It includes the single CPU to execute the program.	It uses multiple tasks for the task allocation.
Concept of Context Switching is used.	Concept of Context Switching and Time Sharing is used.
Switching occurs when the process currently running stops.	Switching occurs after the time slice of the current process is finished.
Multi-programming increases CPU utilization by organizing jobs .	In multi-tasking also increases CPU utilization and it also increases responsiveness.
The idea is to reduce the CPU idle time for as long as possible.	The idea is to further extend the CPU Utilization concept by increasing responsiveness Time Sharing.
It uses job scheduling algorithms so that more than one program can run at the same time.	Time sharing mechanism is used so that multiple tasks can run at the same time.
Execution of process takes more time.	Execution of process takes less time.
Throughput is less	Throughput is moderate
Less Efficiency	Moderate Efficiency

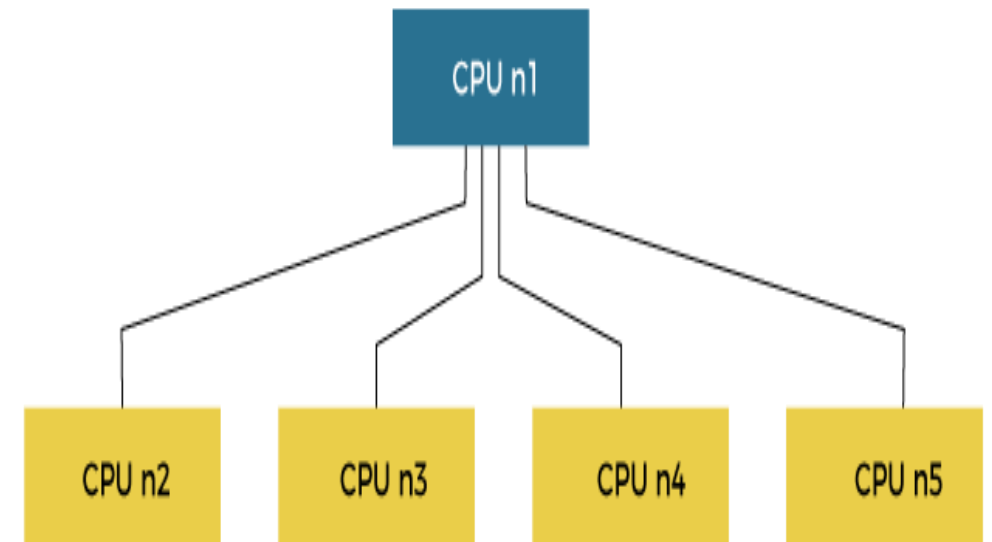
Multiprocessing Operating System

- Multiprocessor operating systems are used in operating systems to boost the performance of **multiple CPUs** *within a single* computer system.
- Multiple CPUs are linked together so that a job can be divided and executed more quickly. When a job is completed, the results from all CPUs are compiled to provide the final output.



Asymmetric Multiprocessing Operating System

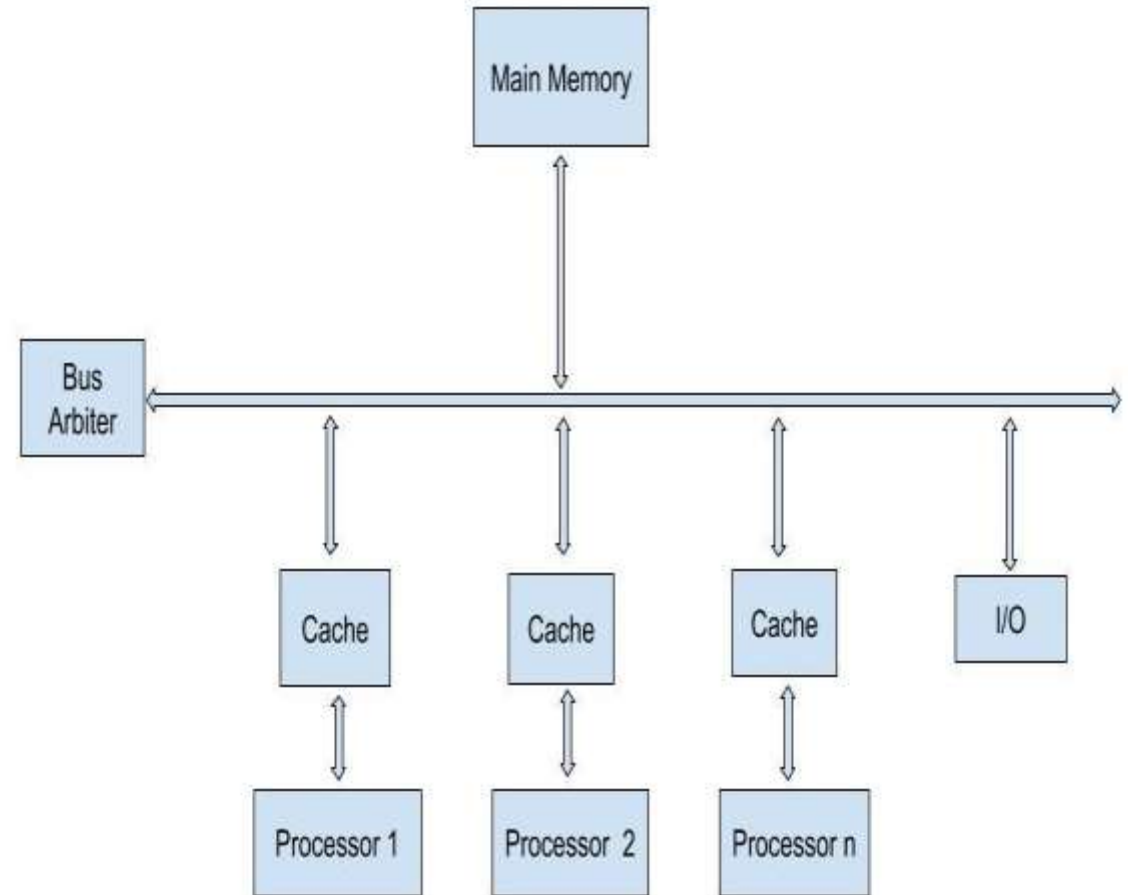
- It has a master slave relationship between the processors.
- **For example**, we have a math co-processor that can handle mathematical jobs better than the main [CPU](#). Similarly, we have an MMX processor that is built to handle multimedia-related jobs.
- Similarly, we have a graphics processor to handle the graphics-related job better than the main processor. When a user submits a new job, the OS has to decide which processor can perform it better, and then that processor is assigned that newly arrived job.
- This processor acts as the master and controls the system. All other processors look for masters for instructions or have predefined tasks. It is the responsibility of the master to allocate work to other processors.



Asymmetric Multiprocessor System

Symmetric Multiprocessing OS

- **SMP** i.e. **symmetric multiprocessing**, refers to the computer architecture where multiple identical processors are interconnected to a single shared main memory, with full accessibility to all the I/O devices, unlike asymmetric MP. In other words, all the processors have common shared(common) memory and same data path or I/O bus as shown in the figure.



Main memory and data bus or I/O bus being shared among multiple processors in SMP

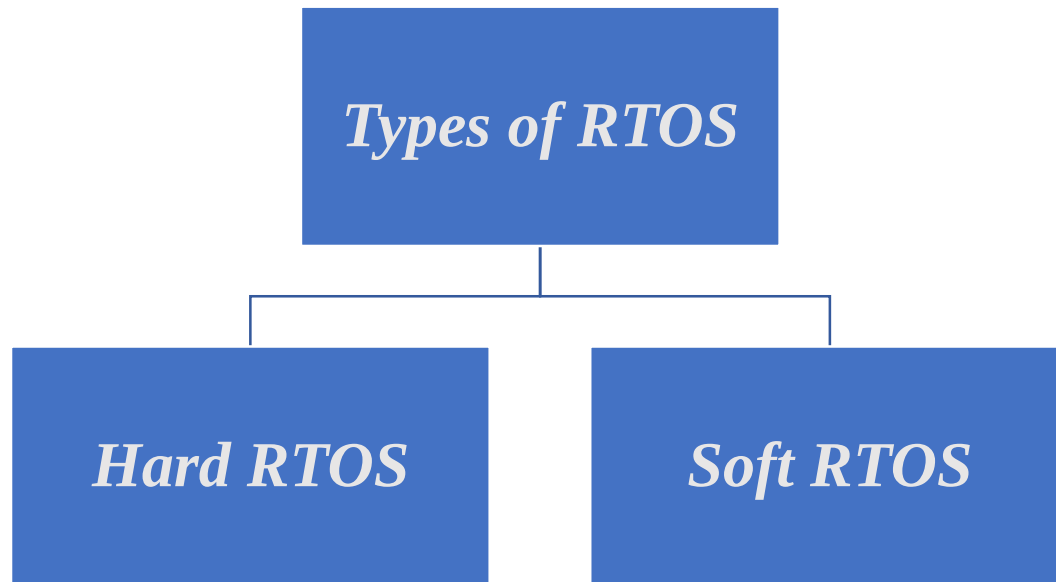
Realtime OS

- A real-time operating system (RTOS) is specialized software designed to process data and respond to events within strict, predictable time limits.

OR

- The real-time operating system used for a real-time application means for those applications where data processing should be done in the fixed and small quantum of time.

Realtime OS



- **Applications:**

- Flight Control System
- Simulations
- Industrial control
- Military applications

Hard RTOS

- These operating systems guarantee that critical tasks are ***completed within a range of time.***
- For example, a robot is hired to weld a car body. If the robot welds too early or too late, the car cannot be sold, so it is a hard real-time system that requires complete car welding by the robot hardly on time.,
- scientific experiments, medical imaging systems, industrial control systems, weapon systems, robots, air traffic control systems, etc.

Soft RTOS

- This operating system provides *some relaxation in the time limit*.
- For example – Multimedia systems, digital audio systems, etc.
- Explicit, programmer-defined, and controlled processes are encountered in real-time systems. A separate process is changed by handling a single external event. The process is activated upon the occurrence of the related event signaled by an interrupt.

Mobile OS



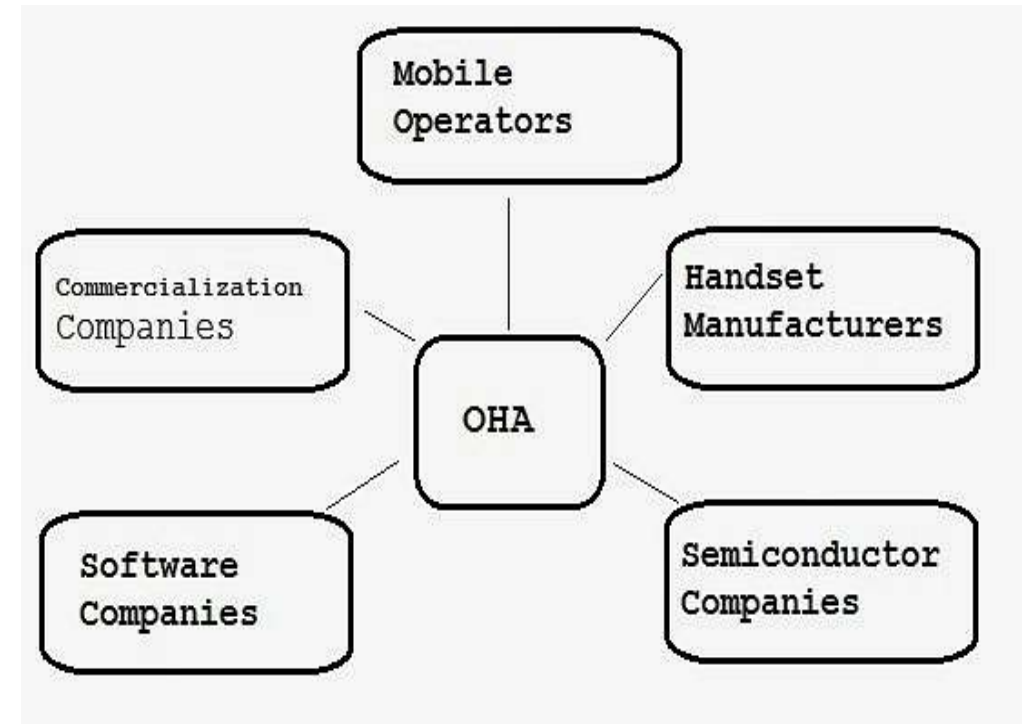
Introduction to Android

- Android is an **open source** and **Linux-based** Operating System .
- It is designed primarily for touch screens mobile devices such as smartphones, tablet computers.
- Android offers a unified approach to application development for mobile devices which means developers need only develop for Android, and their applications should be able to run on different devices powered by Android.
- Android was developed by the Open Handset Alliance, led by Google, and other companies.

Open Handset Alliance (OHA)

Q. Define OHA and state goal of OHA.

- OHA was formed in Nov 2007, created by Google, ***alliance of 84 firms***, for developing open standard platform for mobile devices.
- The Open Handset Alliance (OHA) is a business alliance that was created for the purpose of developing open mobile device standards.
- The OHA has approximately 80 member companies, including HTC, Dell, Intel, Motorola, Qualcomm and Google.
- The members of OHA includes:



Features of Android

1. User Interface
2. Multiple Language Support
3. Multi-tasking
4. Connectivity
5. Extensive Application Support

Describe the different features of Android.

W22, 4m

- **User Interface:** The user interface of the Android operating system is straight forward, and these features make it very user friendly.
- **Multiple Language Support:** Android supports multiple languages in its operating system and one can change the language very easily based on one's requirement, the International languages supported are English, Germany, Chinese, Dutch, French, German, Japanese, Korean, Russian, and many more also some native language of India is also Supported Like Hindi, Marathi, Gujarati, Punjabi and many more.
- **Multi-tasking:** Android provides support to run apps and services in the background with ease which allows the users to use multiple apps at the same time.
- **Connectivity:** Android has extensive support to the connectivity, and it supports connectivity such as WiFi, Bluetooth, Hotspot, CDMA, GSM, NFC, VOLTE, UBB, VPN, 3G network band, and 4G Network Band.
- **Extensive Application Support:** Android have Play store which is used as the major tool to download and update applications on the operating system, however, one can download the installer(often called as APK file) and install it manually, but it is not much recommended as third-party applications could be prone to some security breach in the smartphones.

Features of Android

5. Messaging

SMS and MMS

6. Web browser

Based on the open-source WebKit layout engine, coupled with Chrome's V8 JavaScript engine supporting HTML5 and CSS3.

7. Multi-touch

Android has native support for multi-touch which was initially made available in handsets such as the HTC Hero.

8. Multi-tasking

User can jump from one task to another and same time various application can run simultaneously.

Features of Android

9. Resizable widgets

Widgets are resizable, so users can expand them to show more content or shrink them to save space.

10. Multi-Language

Supports single direction and bi-directional text.

11. GCM

Google Cloud Messaging (GCM) is a service that lets developers send short message data to their users on Android devices, without needing a proprietary sync solution.

12. Wi-Fi Direct

A technology that lets apps discover and pair directly, over a high-bandwidth peer-to-peer connection.

Command Line Interface (CLI)

- A command line interface (CLI) is a *text-based interface* where you can input commands that interact with a computer's operating system. The CLI operates with the help of the default shell, which is between the operating system and the user.
- **CLI shell**
- A *shell* refers to a program that is an intermediary between the user and the operating system. Examples include Bash, cmd.exe, PowerShell, and others for more specific functions.

Command Line Interface (CLI)

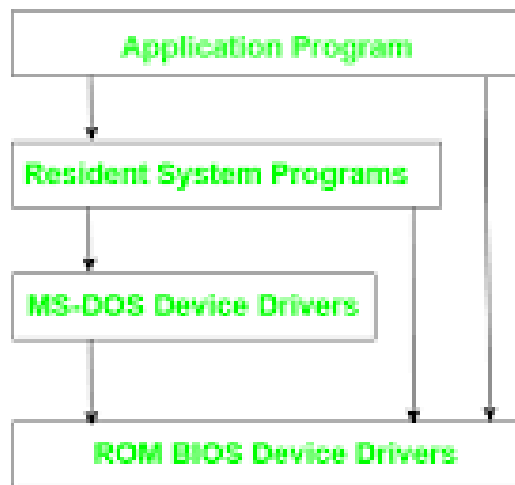
- The shell is responsible for handling various tasks such as command parsing, environment management, and process execution. Shells also support features like these:
- ***Command history***, where you can access previously entered commands using arrow keys or other shortcuts
- ***Input and output redirection***
- ***Piping***—where you connect the output of one command as the input of another—and environment variable management
- Additionally, you can ***customize your shell*** environment. To do this, you can set environment variables, define aliases (shortcuts for longer commands), and create shell scripts for automation or repeated tasks.

MS - DOS

- MS-DOS stands for **Microsoft Disk Operating System**.
- It is an operating system developed by Microsoft for x86 architecture-based computers. MS-DOS is purely a command-line operating system made for IBM-compatible PCs that evolved from 86-DOS.
- MS-DOS was created by Tim Paterson and first released in August 1981 by Microsoft, and it was last upgraded in 1994 with the release of MS-DOS 6.22.
- MS-DOS occupies a **small size** in the hardware memory, so multi-tasking and multi-users are not possible.
- MS-DOS has a **file system** that allows you to organize, read, and write data on your hard drive through the command-line interface.

MS - DOS

MS-DOS layer Structure



```
Current date is Tue 1-01-1980
Enter new date:
Current time is 21:35:24.18
Enter new time:
```

```
The IBM Personal Computer DOS
Version 2.00 (C)Copyright IBM Corp 1981, 1982, 1983
```

```
A>dir
```

```
Volume in drive A has no label
Directory of A:\
```

COMMAND	COM	17664	3-08-83	12:00p
FORMAT	COM	6016	3-08-83	12:00p
CHKDSK	COM	6400	3-08-83	12:00p
SYS	COM	1408	3-08-83	12:00p
DEBUG	COM	11904	3-08-83	12:00p
SLOOP		32	1-01-80	7:44p
		6 File(s)	292864 bytes free	

```
A>_
```

UNIX

- UNIX is an innovative or groundbreaking operating system which was developed in the 1970s by Ken Thompson, Dennis Ritchie.
- It is like a backbone for many modern operating systems like Ubuntu, Solaris, Kali Linux.
- The basic design philosophy of UNIX is to provide simple, powerful tools that can be combined to perform complex tasks. It features a command-line interface that allows users to interact with the system through a series of commands, rather than through a graphical user interface (GUI).

Activities

Terminal

Thu 16:04

raju@raju-HP-Laptop-15-bs1xx: ~/rc_c

File Edit View Search Terminal Help

raju@raju-HP-Laptop-15-bs1xx:~\$ ls

a.out compiler denoising.c Documents examples.desktop lena_rotate.bmp me.txt op.c Public rc_c rc_sm rtlwifi_new Templates threadadd.c xyz.c

cn datasets Desktop Downloads lena.bmp matchanal.py Music Pictures qw.c rc_hw rc_ts snap thread1.bmp Videos

raju@raju-HP-Laptop-15-bs1xx:~\$ cd Downloads

raju@raju-HP-Laptop-15-bs1xx:~/Downloads\$ ls

30740779_1219877821476758_1154882784242696192_n.jpg datasets lena.bmp salt_n_pepper.bmp se05

balloons_noisy.png lena1.bmp saltandpeppernoise.jpeg se04 xampp-linux-x64-7.3.4-0-installer.run

raju@raju-HP-Laptop-15-bs1xx:~/Downloads\$ cd ..

raju@raju-HP-Laptop-15-bs1xx:~\$ cd rc_c

raju@raju-HP-Laptop-15-bs1xx:~/rc_c\$ ls

9march1.cpp a.out b2d.cpp countwords.cpp d2b_part2.cpp homework_ds possible_combinations_backtracking.cpp sumdig.cpp swaparray.cpp

9march2.cpp ap.cpp complementtriangle.cpp d2b.cpp gp.cpp largest_island.cpp reverse_words_in_sentence.cpp sumneg.cpp triangle.cpp

raju@raju-HP-Laptop-15-bs1xx:~/rc_c\$ ls -l

total 92

-rw-r--r-- 1 raju raju 162 Mar 9 17:09 9march1.cpp

-rw-r--r-- 1 raju raju 237 Mar 9 17:15 9march2.cpp

-rwxr-xr-x 1 raju raju 23008 Apr 19 22:15 a.out

-rw-r--r-- 1 raju raju 216 Mar 9 17:40 ap.cpp

-rw-r--r-- 1 raju raju 341 Apr 19 20:48 b2d.cpp

-rw-rw-r-- 1 raju raju 416 Mar 9 17:29 complementtriangle.cpp

-rw-r--r-- 1 raju raju 186 Mar 10 02:32 countwords.cpp

-rw-r--r-- 1 raju raju 253 Apr 19 19:56 d2b.cpp

-rw-r--r-- 1 raju raju 282 Apr 19 20:10 d2b_part2.cpp

-rw-rw-r-- 1 raju raju 216 Mar 9 17:38 gp.cpp

drwxr-xr-x 2 raju raju 4096 Apr 13 18:19 homework_ds

-rw-r--r-- 1 raju raju 638 Apr 19 22:15 largest_island.cpp

-rw-r--r-- 1 raju raju 41 Apr 19 21:24 possible_combinations_backtracking.cpp

-rw-r--r-- 1 raju raju 465 Apr 19 21:18 reverse_words_in_sentence.cpp

-rw-rw-r-- 1 raju raju 229 Mar 9 17:52 sumdig.cpp

-rw-rw-r-- 1 raju raju 303 Mar 9 18:04 sumneg.cpp

-rw-r--r-- 1 raju raju 394 Mar 9 18:15 swaparray.cpp

-rw-r--r-- 1 raju raju 273 Mar 9 17:21 triangle.cpp

raju@raju-HP-Laptop-15-bs1xx:~/rc_c\$

Graphical user interface (GUI)

- GUI lets a user interact with the device/system with the help of graphical elements, like windows, menus, icons, etc. The CLI, on the other hand, lets a user interact with their device/system with the help of various commands. Some OS provide their users with only CLI, while some offer both CLI and GUI.
- Eg:
- Linux
- Windows

What is Linux?

-  Free and open-source OS
-  Used in cloud, DevOps & cybersecurity
-  Fast, secure, and customizable
-  Runs on servers, mobiles & IoT devices



Linux

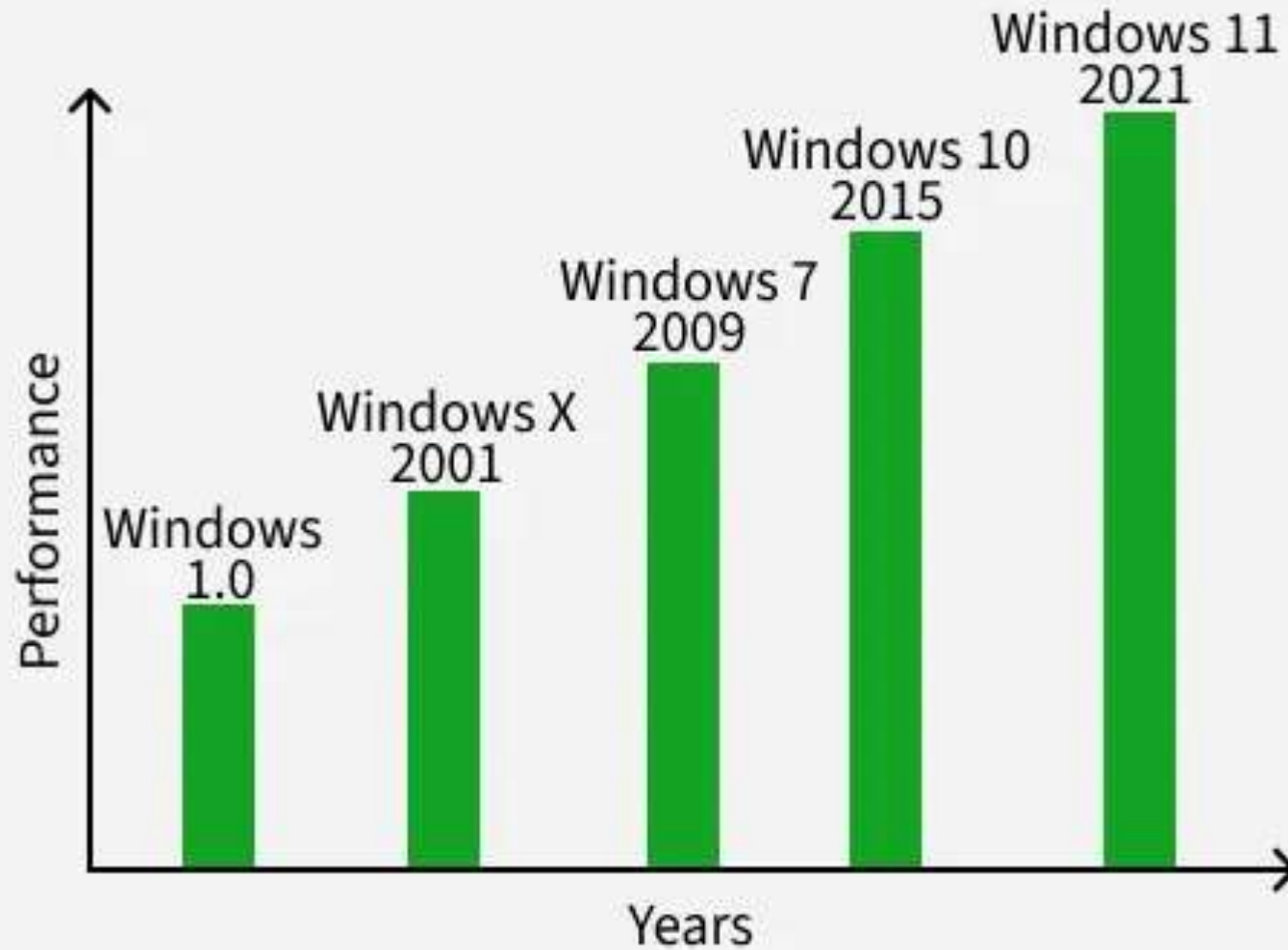
Linux is based on the UNIX operating system, a multi-user and multitasking platform developed in the 1970s at AT&T Bell Labs that became the foundation for many modern operating systems. Linux is free and open source, encouraging global collaboration and innovation while providing strong security, flexibility, and reliable performance across different devices and industries.

- Provides strong security and stability, making it suitable for servers and development environments.
- Allows users to freely use, modify, and distribute the operating system.
- Supports customization for different user requirements and industry applications.
- Backed by a large global community and a wide range of software tools.

Windows OS

Windows is the most popular operating system in the world. It was developed by Microsoft Corporation in 1985 and is also known as Microsoft Windows. Windows has dominated the Operating system market with around 75% of the total stake. The Windows operating system provides many features due to which its popularity has increased exponentially. The features include a very good and interactive graphical user interface (GUI), multitasking, virtual memory management, and support for their devices. Microsoft Windows is not just providing the operating system for computers, but it also offers operating systems for servers and mobile devices. There are many versions of the Windows operating system.

Evolution of Windows



Parameter	Command Line Interface(CLI)	Graphic User Interface(GUI)
Definition	Interaction is by typing commands	Interaction with devices is by graphics and visual components and icons
Understanding	Commands need to be memorized	Visual indicators and icons are easy to understand
Memory	Less memory is required for storage	More memory is required as visual components are involved.
Working Speed	Use of keyboard for commands makes CLI quicker.	Use of mouse for interaction makes it slow
Resources used	Only keyboard	Mouse and keyboard both can be used
Accuracy	High	Comparatively low
Flexibility	Command line interface does not change, remains same over time	Structure and design can change with updates

Different Services of Operating System

Operating System (OS) services are the helpful functions and environment an operating system provides to users and programs to make computer use easy, safe, and efficient.



Key User Services:

- **User Interface (UI):** Provides a screen or command line to let you talk to the computer.
- **Program Execution:** Loads programs into memory and runs them.
- **I/O Operations:** Handles input and output for devices like keyboards and screens.
- **File System:** Lets you create, read, write, and delete files.
- **Communication:** Allows data sharing between different processes or networks.
- **Error Detection:** Finds and fixes hardware or software problems to keep the system stable.

Key System Services:

- **Resource Allocation :** Shares the CPU, memory, and devices fairly among tasks.
- **Accounting :** Keeps track of how system resources are being used by users and programs.
- **Memory Management :** Manages the allocation and deallocation of memory to programs.
- **Security and Protection :** Keeps user data safe and blocks unauthorized access.

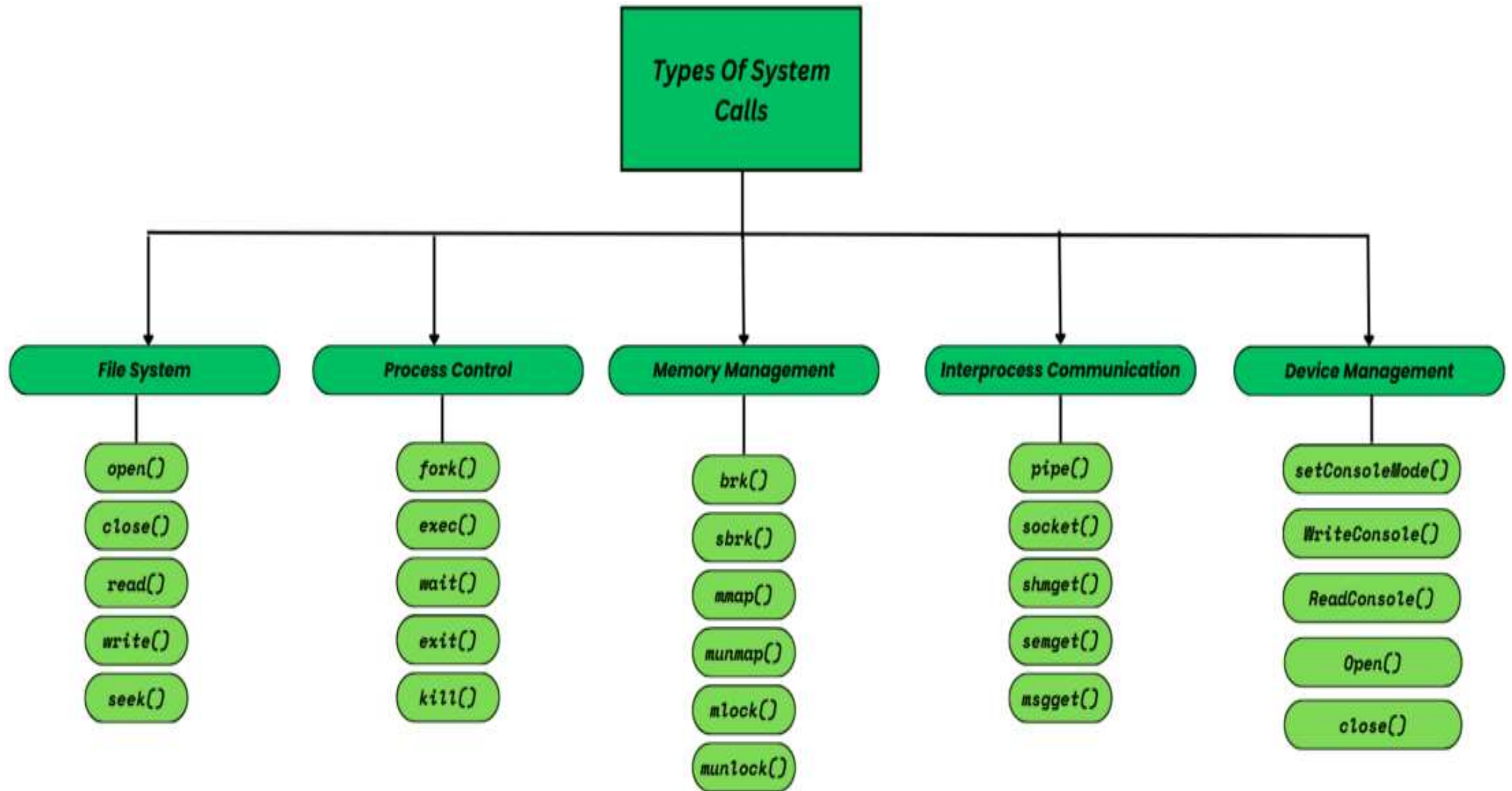
System Calls

A system call is a programmed request where a running user program asks the operating system's kernel to perform a restricted task, such as opening a file, creating a process, or using hardware. It acts as a secure bridge between user applications and system hardware.

Kernel: A kernel is the core part of an operating system. It acts as a bridge between software applications and the hardware of a computer.

Types of system calls

- **File System:** Used to create, open, read, write, and manage files and directories.
- **Process Control:** Used to create, execute, coordinate, and terminate processes.
- **Memory Management:** Used to allocate, deallocate, and manage memory for processes.
- **Interprocess Communication (IPC):** Used for data exchange and communication between different processes.
- **Device Management:** Used to request and release devices, and to perform read/write operations on them.



The image features a white background with decorative geometric shapes in the corners. In the top-left corner, there are overlapping blue squares. In the top-right corner, there are overlapping yellow squares. In the bottom-right corner, there are overlapping blue squares. The text "Thank You!" is centered in a black, cursive font.

Thank You!

References:

- https://youtu.be/75z6HRmo_CU
- <https://www.geeksforgeeks.org/operating-systems/ms-dos-operating-system/>